3-Level Geometric Modeler: CSG, Subdivision Trees and Boundary Representation

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Abstract

Geometric modeling embraces computational geometry and extends beyond this to the newer field of solid modeling,creating an elegant sythesis of geometry and the computer. Solid modeling is distinguished from related areas of geometric modeling and computer graemptysetcs by its emphasis on physical fidelity. Together, the principles of geometric and solid modeling form the foundation of computer-aided design and in general support the creation, exchange, visualization, animation, interrogation, and annotation of digital models of physical objects.

I. Introduction

Geometric modeling embraces computational geometry and extends beyond this to the newer field of solid modeling,creating an elegant sythesis of geometry and the computer. Solid modeling is distinguished from related areas of geometric modeling and computer graemptysetcs by its emphasis on physical fidelity. Together, the principles of geometric and solid modeling form the foundation of computer-aided design and in general support the creation, exchange, visualization, animation, interrogation, and annotation of digital models of physical objects.

II. Definitions

2.1 Regularized Set

A set $S \subset \mathbb{R}^3$ is a regularized set iff $S = \overline{int(S)}$.

2.2 Solid

Our goal is mainly focused on solids. A solid S is defined as a regularized subset of $\mathbb{R}^3$ ($S \subset \mathbb{R}^3$)

2.3 Polyhedron

A polyhedron P is the intersection of several half spaces. And the half spaces can be descirbed as a set $H = \{h_1, h_2, h_3, ...h_n\}$

A polyhedron P contains features which can be described as three sets: a Vertex Set $V = \{v_1, v_2, ..., v_m\}$ in which $v_i (1 \leq i \leq m)$ represents a vertex of the polyhedron P, $E = \{e_1, e_2, ..., e_n\}$ in which $e_j (1 \leq j \leq n)$ represents an edge of P, $F = \{f_1, f_2, ..., f_p\}$ in which $f_k (1 \leq k \leq p)$ represents a face of P.

III. CSG objects

Just as a polyhedron can be described as the intersection of half spaces, a CSG (Constructive Solid Geometry) model describes a polyhedron as Boolean operation on primitives which are the simplest solid objects used for the representation. The Boolean operations on primitives are intersection, union, difference and complement. The primitives are solids. Suppose the equation of a surface is $f(x, y, z) = 0$, the primitive decided by this surface are either $f(x, y, z) \leq 0$ or $f(x, y, z) \geq 0$. For a polyhedron we define primitives as half spaces. But in more complicated cases, we need complex primitives like a cylinder, a sphere and so on.

3.1 Primitives:

These primitives comes from semigeometric definition and all these primitives are subsets of $\mathbb{R}^3$.

1. Half Space: A half space H is a subset of $\mathbb{R}^3$, H is defined by an inequation. Suppose the equation of a plane in $\mathbb{R}^3$ is $f(x,y,z) = a*x+b*y+c*z-d = 0$,which is a linear function of coordinates(x,y,z), then the upper half space defined by equation $f(x,y,z) = 0$ are $f(x,y,z) = a*x+b*y+c*z-d \geq 0$, likewise the lower half space defined by equation $f(x,y,z) = 0$ are $f(x,y,z) = a*x+b*y+c*z-d \leq 0$.
2. Sphere : A sphere S should also be a subset of $\mathbb{R}^3$. The boundary of the sphere are defined by an equation $f(x,y,z) = (x-x_0)^2+(y-y_0)^2+(z-z_0)^2-r^2 = 0$. When $f(x,y,z) \geq 0$, we define the outer space of the sphere including the boundary. Or when $f(x,y,z) \leq 0$, we define the inner halp space including the boundary.
3. Cylinder : A Cylinder C are a subset of $\mathbb{R}^3$. The boundary of this cylinder are defined by an equation $f(x,y,z) = 0$

IV. Subdivision trees:

1. Box: B is a box iff $B \subset \mathbb{R}^3$, $B = \{(x,y,z)|x_1 \leq x \leq x_2, y_1 \leq y \leq y_2, z_1 \leq z \leq z_2\}$, which is a regularized subset of $\mathbb{R}^3$
2. ? Box of a polyhedron P:
   (a) Empty Box: a box B is an Empty Box iff $B \cap P = \emptyset$.
   (b) Full Box: a box B is a Full Box of Polyhedron P iff $B \cap P = B$
   (c) Vertex Box: a box B is a Vertex Box iff there exists only one vertex $v \in V \cap B$, s.t. $\forall e \in E$, if $e \cap B \neq \emptyset$, v bounds e, and $\forall f \in F$, if $f \cap B \neq \emptyset$, v bounds F.
   (d) Edge Box: a box B is an Edge Box iff $V \cap B = \emptyset$ and there exists only one edge $e \in E$, $e \cap B \neq \emptyset, \forall f \in F$, if $f \cap B \neq \emptyset$, e bounds f.
   (e) Face Box: a box B is a Face Box iff $\forall e \in E$, $e \cap B = \emptyset$, and there exists one face $f \in F$, and $f \cap B \neq \emptyset$.
3. Adequate subdivision tree: In $\mathbb{R}^3$, a subdivision tree are a octree in which each node of the tree would be a finite Box. The root are a finite box B_0. We can construct the tree by beginning with the B_0, and subdividing the tree until all leaves are ? Boxes.